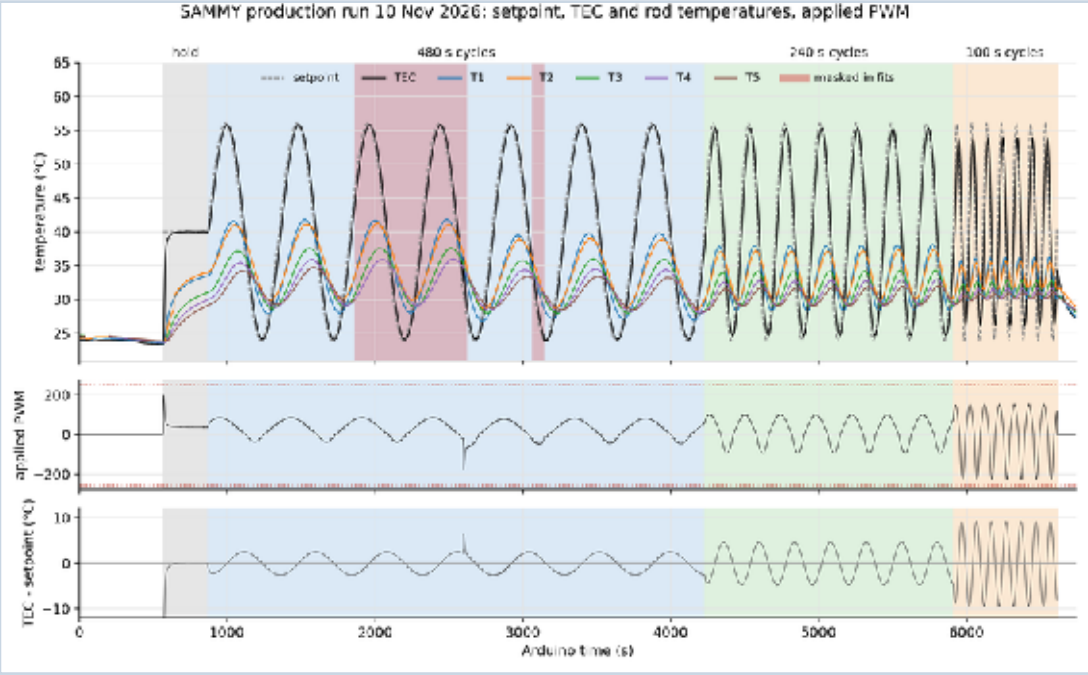
Ångström run: thermal diffusivity of an aluminum rod

Phys 39 · Module 8, Part I · PI-controlled TEC drive at periods of 480, 240 and 100 s

Samuel Cohen (SAMMY copy) · run of 10 November 2026 · sammy_angstrom_live_20261110_164057.csv



Weighted mean of the three periods ($\chi^2 = 3.6$ for 2 degrees of freedom)

$\kappa = 9.12 \pm 0.09 \times 10^{-5}$ m²/s

- 32 % above the handbook value for 6061-T6 (6.9×10^{-5} m²/s). Kapil's PI data on the same rod show the same offset.
- Within 0.5 to 1.7 σ of Kapil's PI fits at the same periods.
- Only 240 s has the five settled cycles the lab asks for; the other periods are supplementary.

The Ångström method in one slide

- Drive one end of a long rod with a sinusoidal temperature of period P ($\omega = 2\pi/P$).
- A damped thermal wave travels down the rod: its amplitude falls as e^(−qx) and its phase lags by q′x.
- Fit each thermistor record with a sine: amplitude B and phase φ at its position x.
- ln B and φ are straight lines in x. Their slopes are −q and −q′.
- The product qq′ fixes the diffusivity. Heat loss from the side of the rod changes q and q′ separately ($q^2 - q'^2 = \nu/\kappa$) but not their product.

temperature along the rod: steady profile plus a travelling wave

$$T(x,t) = T_m(x) + B(x)\cos\left(\omega t + \varphi(x)\right)$$

amplitude decay and phase lag, both linear in x (after taking ln B)

$$B(x) = B_0\, e^{-qx}, \qquad \varphi(x) = \varphi_0 - q'x$$

thermal diffusivity from the product of the two slopes

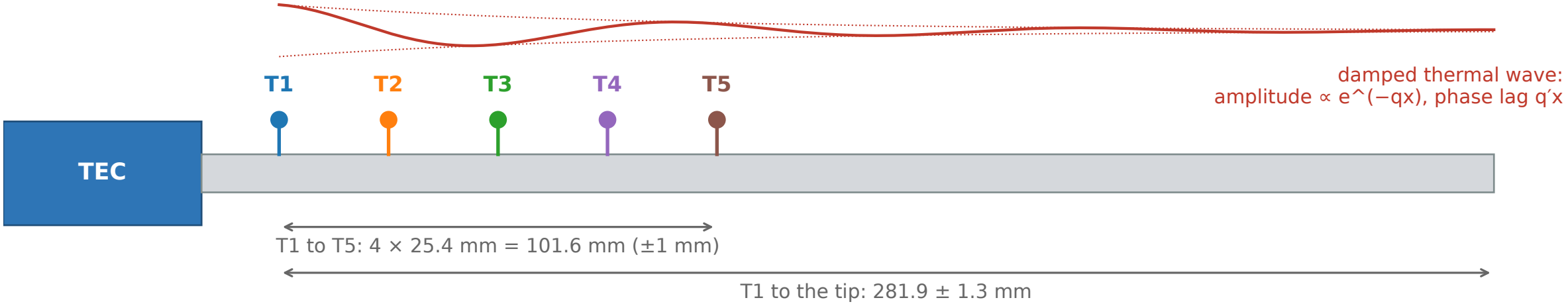
$$\kappa = \frac{\omega}{2\,q\,q'}$$

side-loss rate (the small difference of two nearly equal numbers)

$$\nu = \kappa\,(q^2 - q'^2)$$

Two slopes and one frequency give κ. Only amplitude ratios and phase differences between positions matter, so constant sensor offsets cancel.

Set-up: rod, sensors, controller



- Rod: 6061-T6 aluminum, d = 12.757 ± 0.006 mm.
- Thermistors T1 to T5 at 0, 25.4, 50.8, 76.2 and 101.6 mm (±1 mm); the tip is 281.9 ± 1.3 mm from T1.
- Sensor offsets from the 26 Oct calibration run are applied in the analysis only. They do not change B or φ.
- TEC on an H-bridge driven by an Arduino Uno. PI control of the TEC thermistor (Kp 12, Ki 0.2) with heating-to-cooling ratio r = χh/χc = 2.771.
- Setpoint 40 ± 16 °C sine, computed on the Arduino so period and phase are exact.
- The Arduino latches the drive off if T1 exceeds 56 °C or the TEC thermistor 65 °C. The Python GUI logs 18 columns at about 1.8 records per second.

> The wave is launched by the TEC and measured by five thermistors along 101.6 mm. The sine setpoint and the safety latches run on the Arduino, so the forcing does not depend on the computer's timing.

Procedure: one schedule, three periods

idle	hold	7 × 480 s	7 × 240 s	7 × 100 s	
9.5 min	5.0 min	56.0 min	28.0 min	11.7 min	

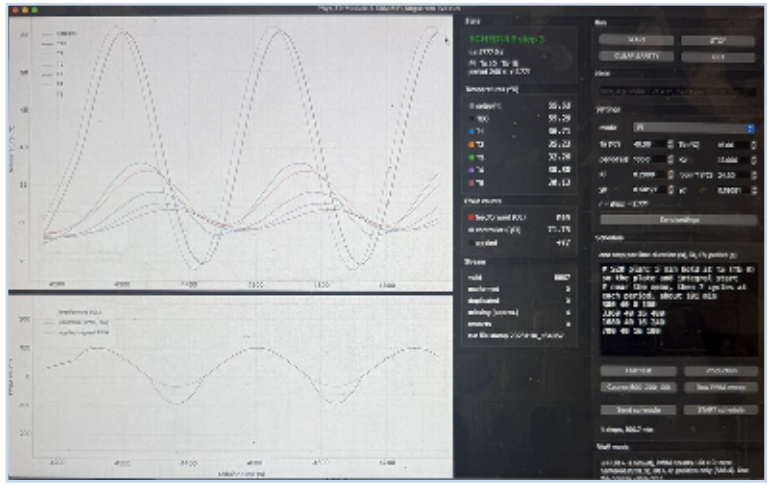
Time axis of the logged file: 112.2 min, 12407 records

Step	What	Setpoint	Period	Planned	Ran
1	Hold at 40 °C to settle plate and integral term	40 °C	none	300 s	5.0 min
2	Oscillation, 7 cycles	40 ± 16 °C	480 s	3360 s	56.0 min
3	Oscillation, 7 cycles	40 ± 16 °C	240 s	1680 s	28.0 min
4	Oscillation, 7 cycles	40 ± 16 °C	100 s	700 s	11.7 min

- 9.5 min elapsed between the start of logging and step 1: lead handling, then idle recording (room temperature 24.3 °C = mean of T1 to T5).
- Seven cycles per period let the start-up transient die out and still leave a settled block to fit.
- The raw CSV is never edited. Masks and sensor offsets are applied only inside the analysis.

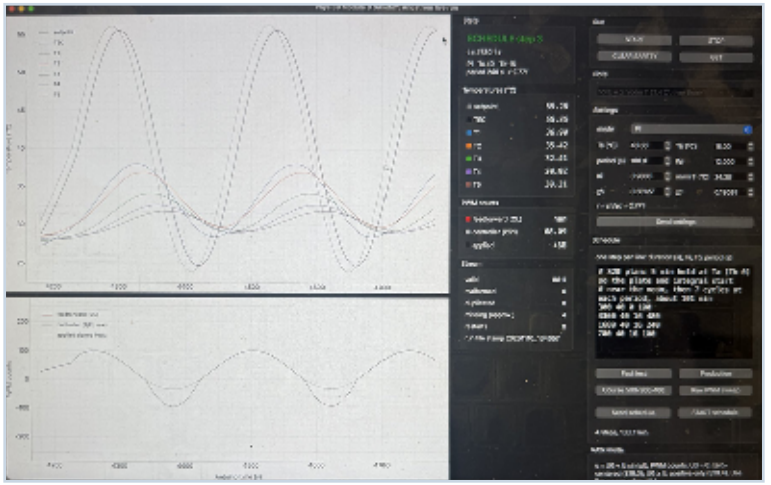
> One uninterrupted schedule gives three periods (480, 240 and 100 s) on the same rod with the same sensors, so the periods can be compared directly.

The run on the bench: live GUI photos



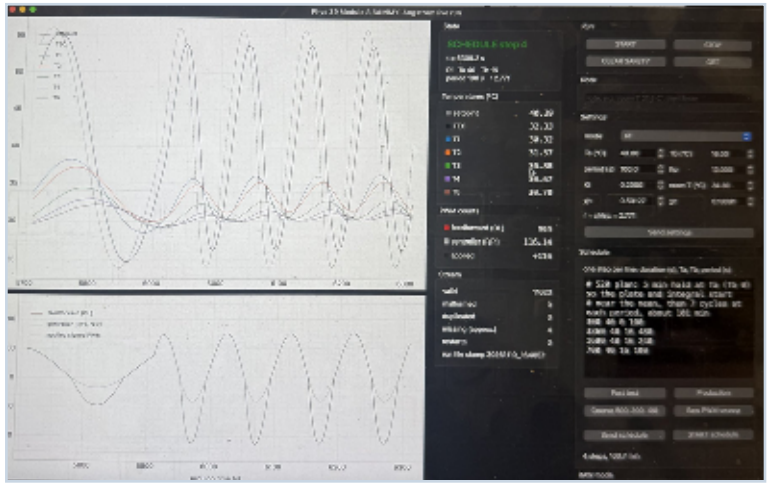
IMG_2550, 6:00 PM: step 3 (240 s period), t = 4777.9 s

On screen: setpoint 55.53 °C, TEC 55.29 °C, T1 36.71, T2 35.23, T3 32.26, T4 30.80, T5 30.13 °C; applied PWM +72 (record 8807).



IMG_2551, 6:00 PM: step 3 (240 s period), t = 4780.1 s

On screen: setpoint 55.28 °C, TEC 55.35 °C, T1 36.90, T2 35.42, T3 32.41, T4 30.92, T5 30.21 °C; applied PWM +68 (record 8811).



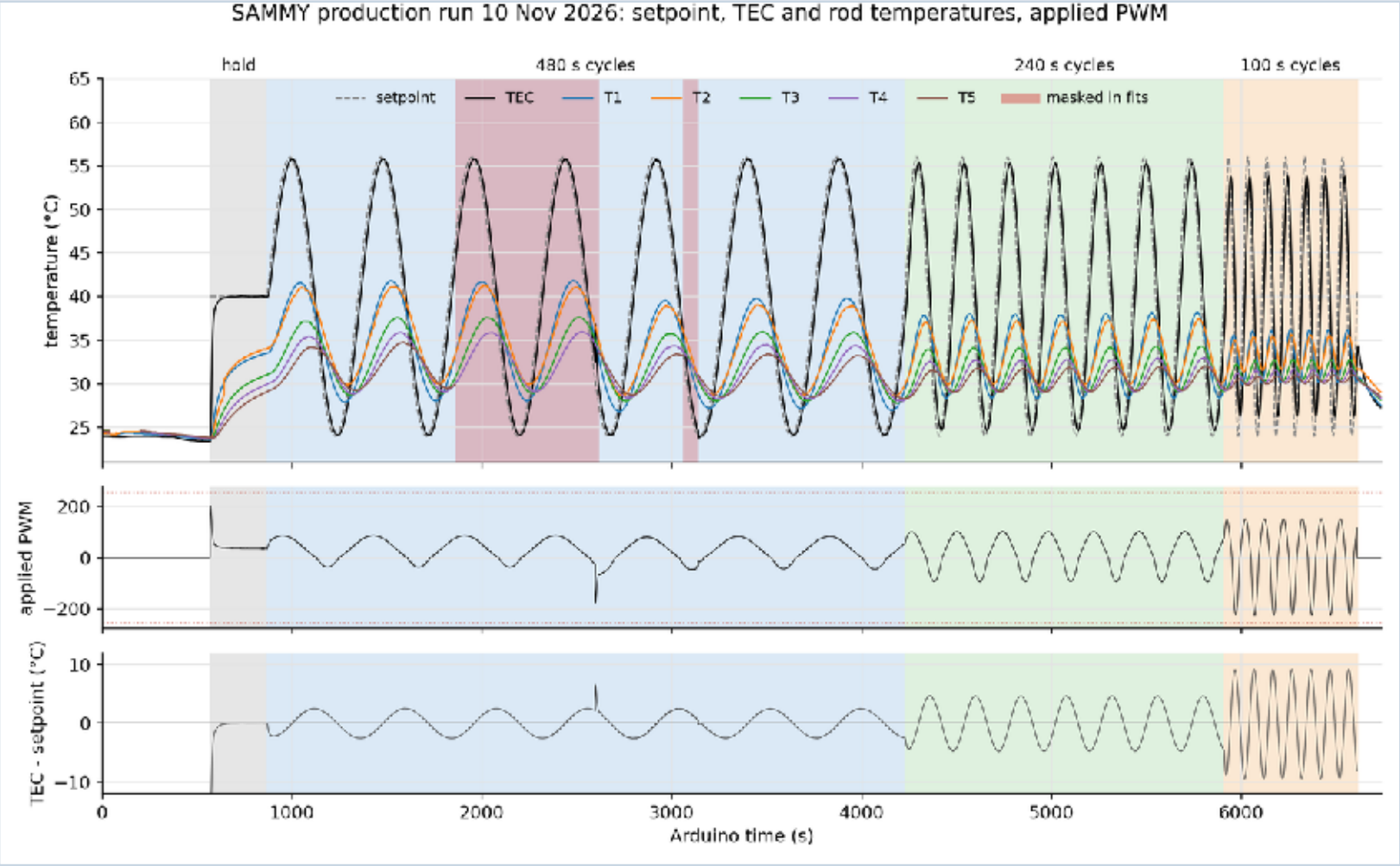
IMG_2558, 6:26 PM: step 4 (100 s period), t = 6309.2 s

On screen: setpoint 40.39 °C, TEC 32.33 °C, T1 30.32, T2 31.57, T3 30.88, T4 30.67, T5 30.70 °C; applied PWM +116 (record 11623).

- Plots on the left (temperatures above, PWM below), live readouts in the middle, schedule and controls on the right.
- 240 s step: the TEC thermistor follows the setpoint near its 55 °C peak while the rod sensors show smaller, later peaks (T1 36.7 °C down to T5 30.1 °C): the wave dying out along the rod.
- 100 s step: the TEC swings quickly and the rod sensors barely move (T1 to T5 within 30.3 to 31.6 °C at this instant).

> The numbers on screen are the logged values (record numbers 8807, 8811 and 11623), so the photos document the run without adding any data of their own.

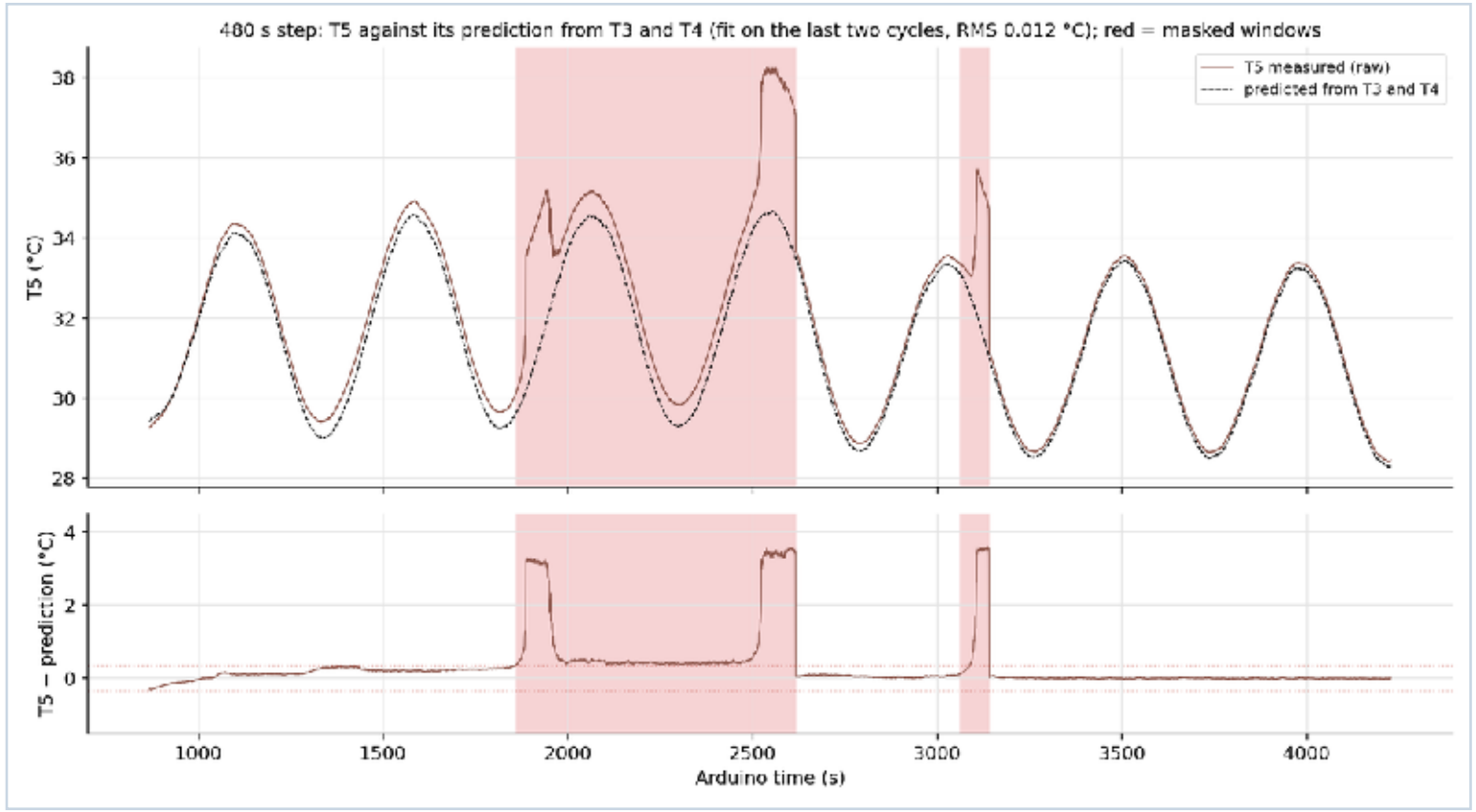
The whole run



- Setpoint (dotted), TEC thermistor (black) and T1 to T5. Shaded bands are the four schedule steps; red marks masked samples.
- All four steps ran to the end (5.0, 56.0, 28.0, 11.7 min).
- PWM never reached ±255 (range −225 to 202).
- T1 peaked at 41.9 °C and the TEC thermistor at 55.9 °C (latch limits 56 and 65 °C).
- 4 gaps longer than 0.75 s, 0 during the schedule; 0 duplicated records and 0 restarts.

The controller held the TEC thermistor on the 40 ± 16 °C sine for 100 minutes without saturating or tripping a latch.

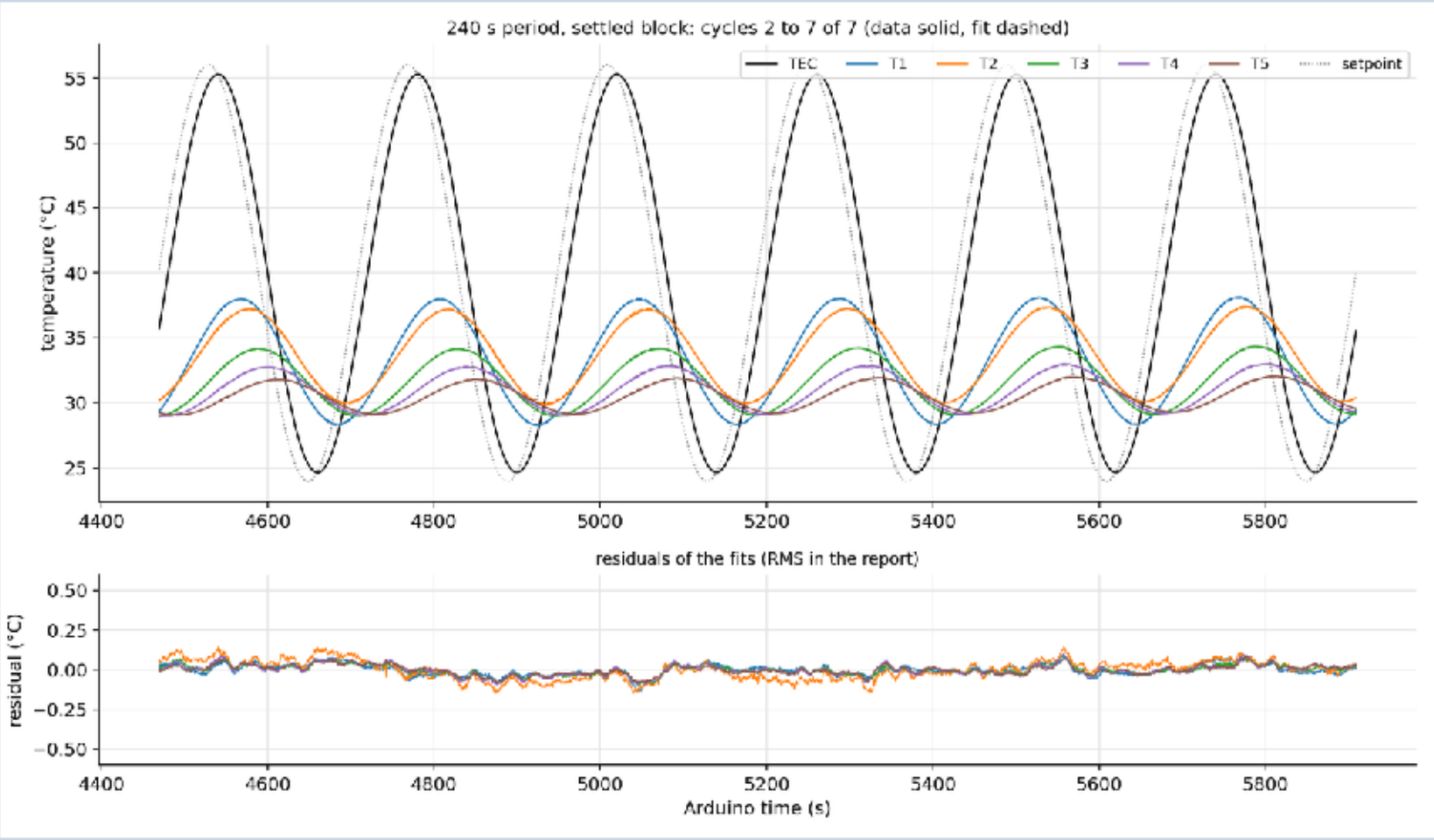
Data quality: what was masked and why



- T3, T4 and T5 spiked while the leads were handled before the run (up to 33, 117 and 102 °C): masked from 36 to 200 s.
- In the 480 s step T5 read 1 to 4 °C above its prediction from T3 and T4 for 1860 to 2618 s and up to 3.6 °C for 3060 to 3141 s: masked.
- The prediction uses T3, T4 and their slopes, fitted on the last two cycles of each step (RMS 0.012 °C).
- Outside the masks T5 stays within 0.35 °C (480 s), 0.14 °C (240 s) and 0.15 °C (100 s) of it.
- The raw CSV is unchanged: 2462 samples are set to nan inside the fits only.

A contact fault on T5 was removed by an explicit, documented mask rather than by editing the data; the mask file sits next to the run CSV.

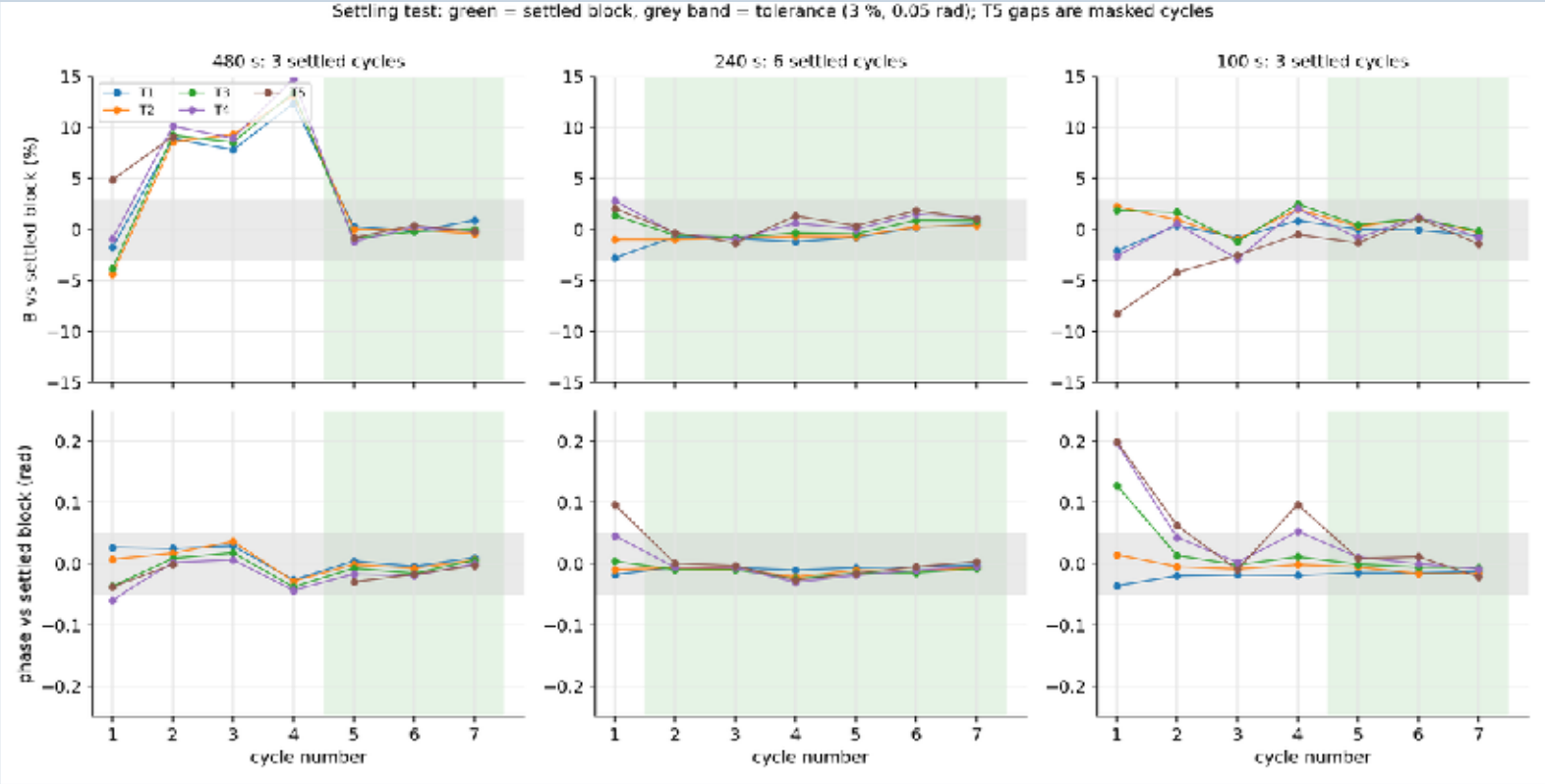
Step 1 of the analysis: fit every cycle



- Each sensor and each cycle is fitted separately: $T = a + B\cos(\omega t + \varphi) + c\,(t - t_{mid})$. The linear term absorbs slow drift.
- Example: the 240 s step. The TEC thermistor follows the setpoint; the rod sensors show smaller and later waves, smaller the further from the TEC.
- Residuals stay within about ±0.15 °C.
- The cycle-to-cycle scatter of B and φ ($sd/\sqrt{N}$) is the uncertainty carried into the spatial fit, as in Kapil's angstrom_fit.py.

A sine fit per cycle turns each temperature record into one amplitude and one phase per sensor, with an error bar taken from the data themselves.

Step 2: which cycles have settled?



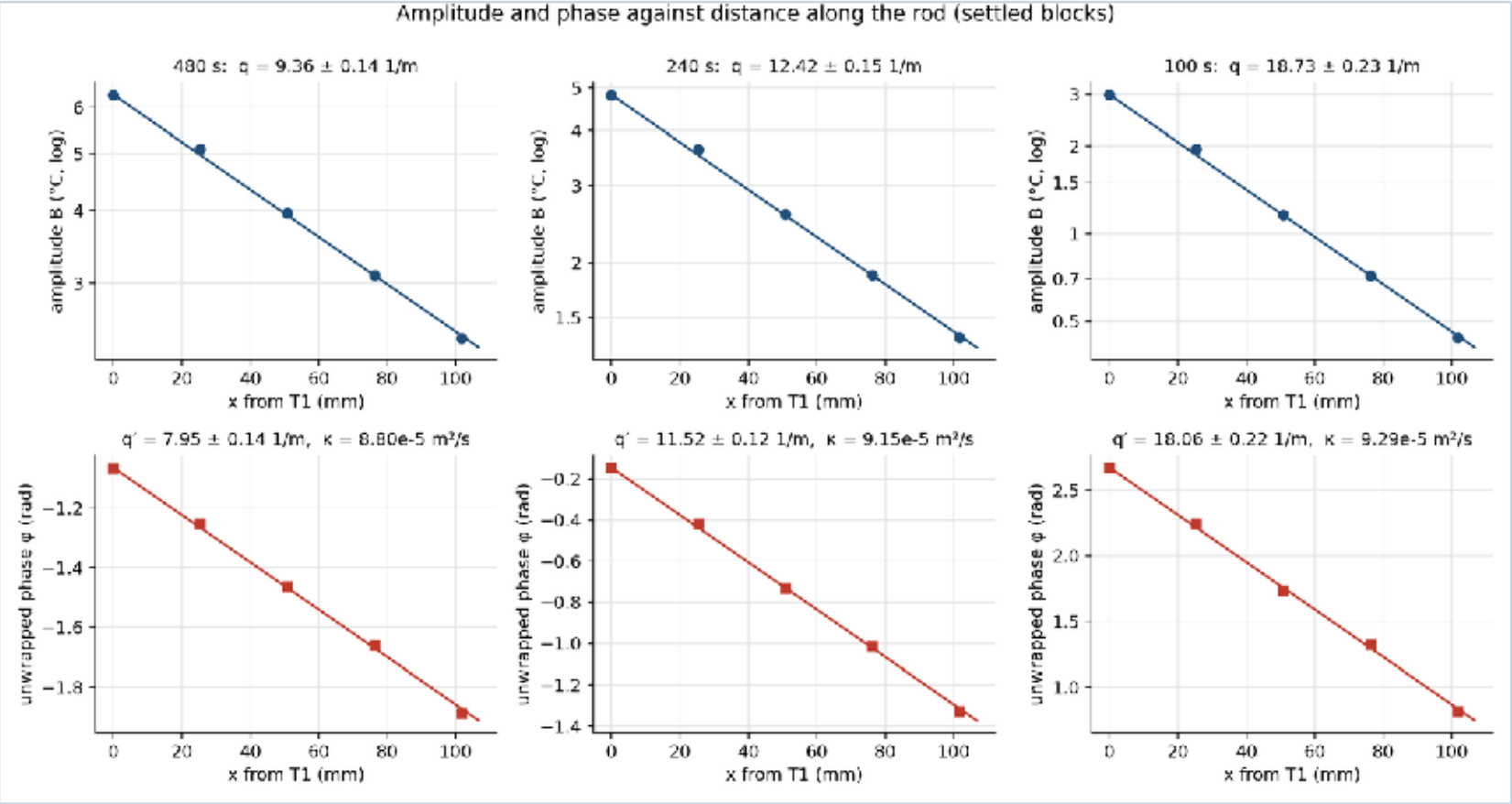
Lines show the change of B (top) and φ (bottom) of every sensor from cycle to cycle, relative to the settled block.

P (s)	Settled cycles	≥ 5
480	5 to 7 (3)	no
240	2 to 7 (6)	yes
100	5 to 7 (3)	no

- A cycle joins the block while every rod sensor is within 3 % in B and 0.05 rad in φ of the block median, counting back from the last cycle.
- 480 s: at t = 2598 s the TEC reading stepped up by 4 °C and all rod amplitudes afterwards were about 10 % lower. The cause is unknown.
- 100 s: T5's amplitude is only 0.44 °C, so 3 % is 0.013 °C.

> Only the 240 s step has five or more settled cycles. The 480 s and 100 s results use three cycles each and are supplementary; the tolerances are chosen for this analysis.

Step 3: amplitude and phase against distance



- Top: ln B against x is a straight line of slope −q. Bottom: the unwrapped phase is a straight line of slope −q′.
- q = 9.4, 12.4, 18.7 1/m at 480, 240, 100 s: shorter periods decay faster.
- Lines are weighted fits; the errors are scaled by √(χ²/dof) when that is above 1 (Kapil's rule).
- q and q′ differ by only 15 % (480 s), 7 % (240 s), 4 % (100 s), so side loss is a small correction to the pure-conduction wave.

Five sensors fall on straight lines in both amplitude and phase, which is what the model predicts; κ = ω/(2qq′) follows from the two slopes.

Results: thermal diffusivity at each period

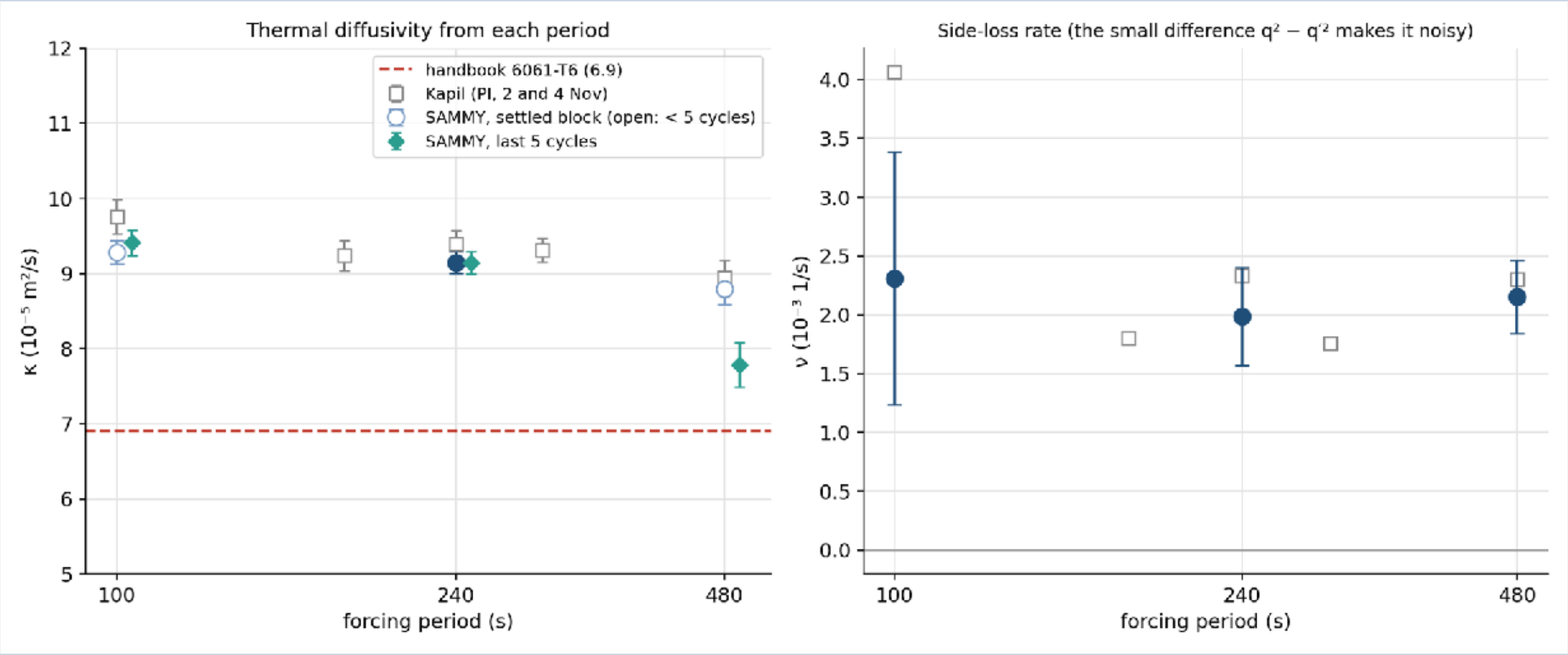
P (s)	Settled	q (1/m)	q′ (1/m)	κ (10^{-5} m²/s)	Kapil κ (PI)	Difference	k (W/m K)
480	3 of 7	9.36 ± 0.14	7.95 ± 0.14	8.80 ± 0.21	8.94 ± 0.24	−0.5 σ	213
240	6 of 7	12.42 ± 0.15	11.52 ± 0.12	9.15 ± 0.14	9.40 ± 0.17	−1.1 σ	221
100	3 of 7	18.73 ± 0.23	18.06 ± 0.22	9.29 ± 0.16	9.76 ± 0.23	−1.7 σ	225

Highlighted row: meets the five-settled-cycle minimum. Kapil's values are his PI fits of 2 and 4 Nov on the same rod; k uses the handbook ρ and cp.

- Weighted mean of the three periods: $9.12 \pm 0.09 \times 10^{-5}$ m²/s ($\chi^2 = 3.6$ for 2 degrees of freedom, so the periods are consistent with one value).
- Handbook 6061-T6 (k = 167 W/(m K), ρ = 2700 kg/m³, cp = 896 J/(kg K)) gives $\kappa = 6.9 \times 10^{-5}$ m²/s: the measurement is 32 % higher. Kapil's five periods are 30 to 41 % higher.
- A spacing error cannot explain this: κ scales with spacing squared, so about 15 % would be needed, while the sensor positions are good to about 1 %.
- Using the last five cycles of every period (Kapil's rule) instead of the settled block changes κ by −11.5 % at 480 s, 0.0 % at 240 s, +1.3 % at 100 s; the 480 s window straddles the disturbance at t = 2598 s.

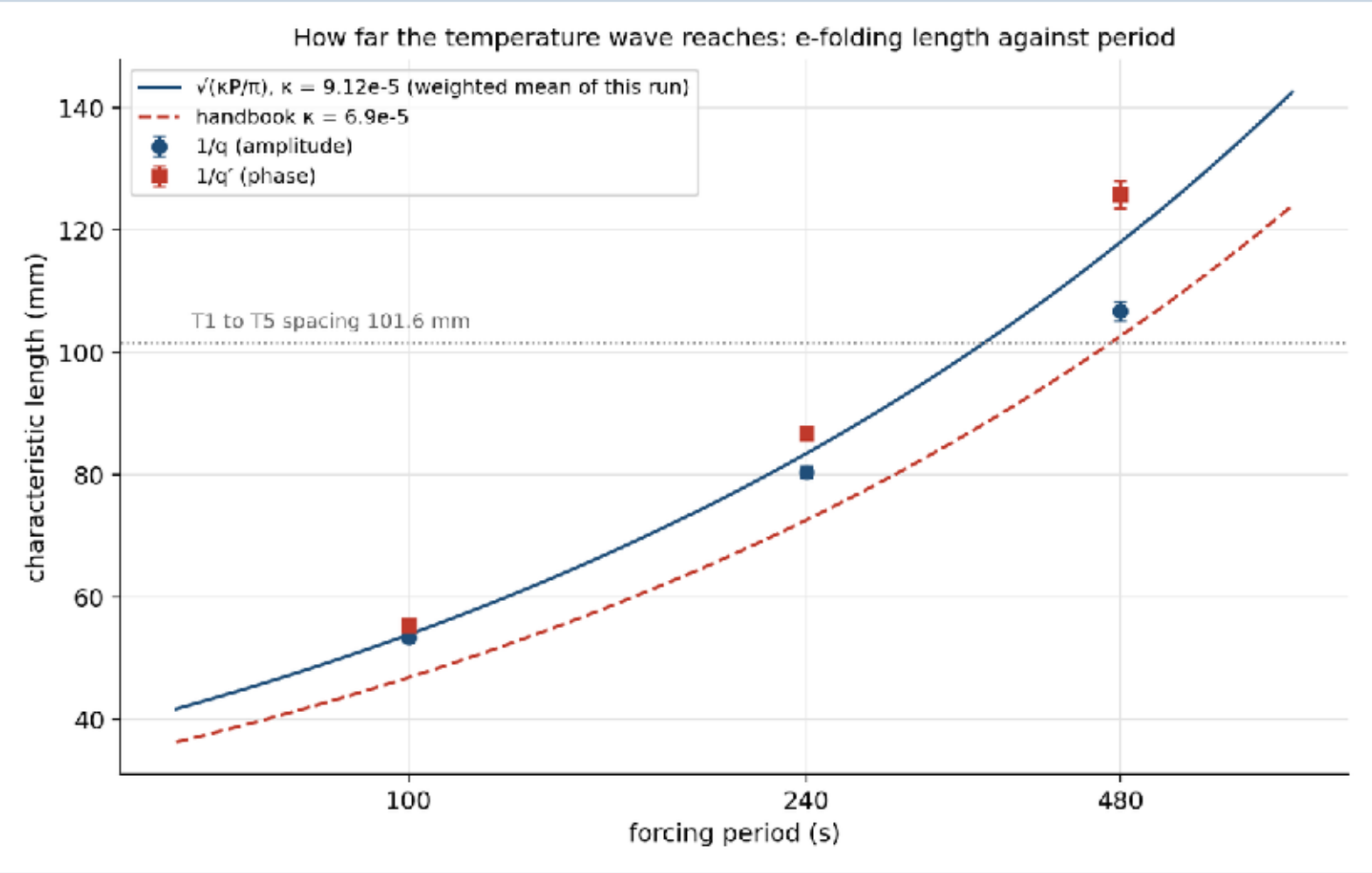
> $\kappa = 9.12 \pm 0.09 \times 10^{-5}$ m²/s, in agreement with Kapil's data but about 32 % above the handbook value; the cause is not resolved by this run.

κ against forcing period



No significant trend with period ($\chi^2 = 3.6$ for 2 dof). Differences from Kapil's PI fits: $-0.5\,\sigma$ at 480 s, $-1.1\,\sigma$ at 240 s, $-1.7\,\sigma$ at 100 s. ν, from the small difference $q^2 - q'^2$, is noisy (right panel).

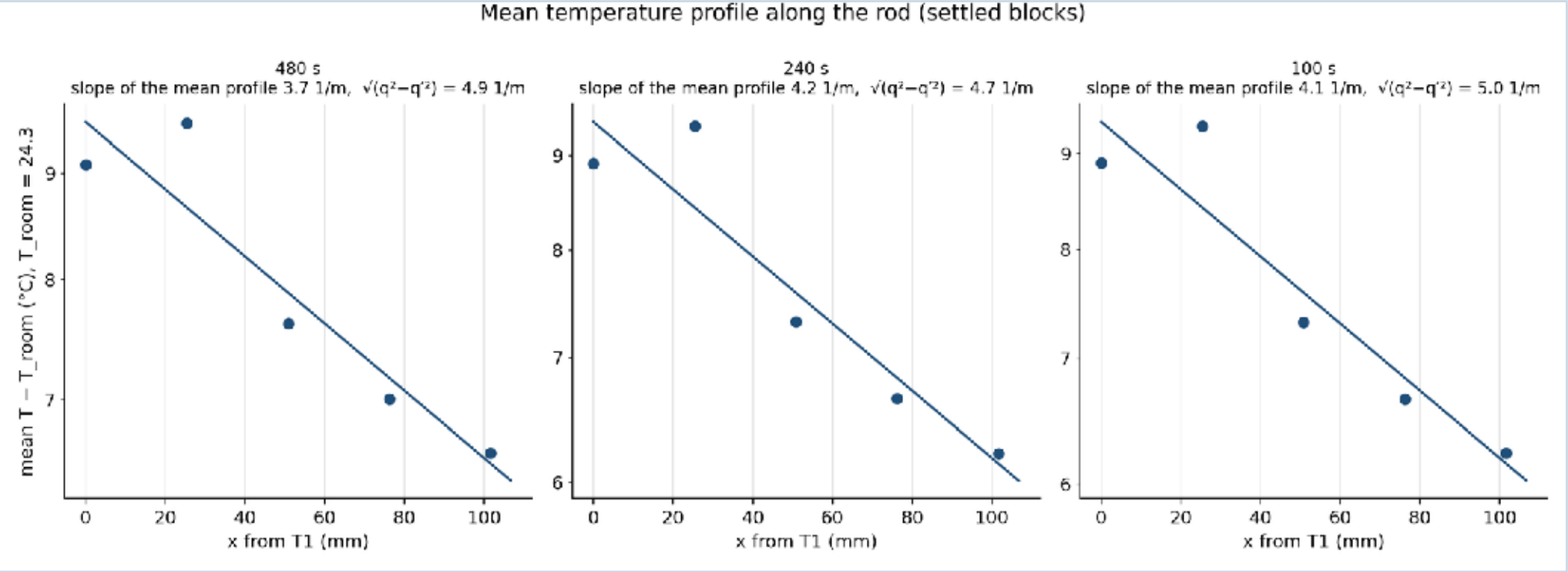
How far does the wave reach?



- 1/q is the distance over which the amplitude falls by a factor e: 107 mm (480 s), 80 mm (240 s), 53 mm (100 s).
- The five sensors span 101.6 mm, so at 100 s the last sensor sees only about 15 % of the amplitude of the first (T5 amplitude 0.44 °C).
- At 480 s the rod is only 2.6 e-folding lengths long, which gives the largest finite-length difference (3.6 %).
- A longer period reaches further but needs more settled cycles; a shorter period is cheap but leaves T5 with a small signal.

The three periods probe the rod at different length scales and give the same κ, which supports using one conduction model for all of them.

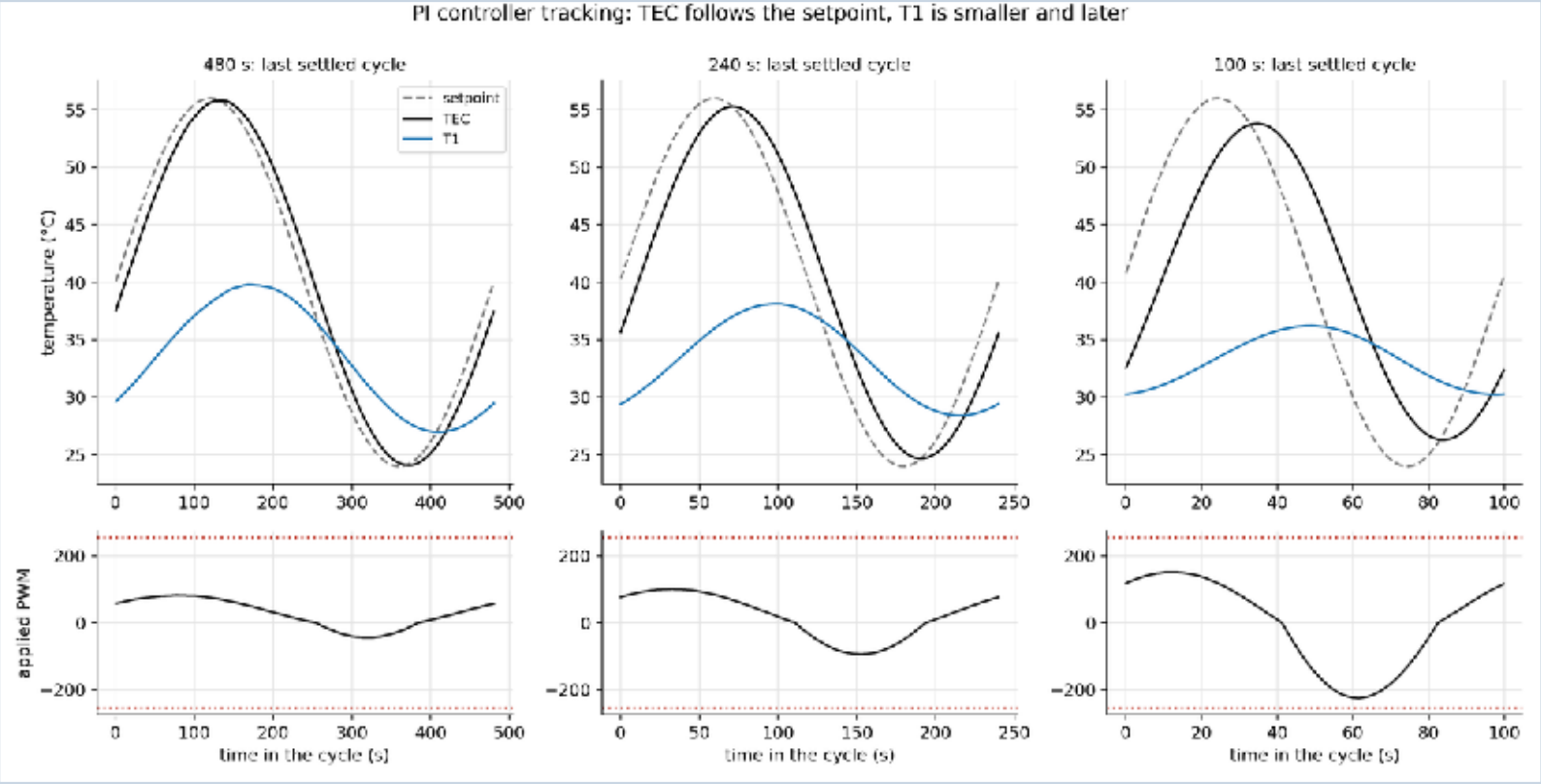
Is the semi-infinite model good enough?



P (s)	Finite-length difference (max)	Radial Biot number	m from √(q²−q′²) (1/m)	m from the mean profile (1/m)
480	3.60 %	2.5e-04	4.95	3.73 ± 0.14
240	1.21 %	2.2e-04	4.66	4.17 ± 0.17
100	0.12 %	2.5e-04	4.99	4.12 ± 0.16

Finite length and radial gradients are negligible. The mean-profile slope does not match √(q²−q′²). T2's mean temperature is 0.4 °C above T1's, which conduction away from the TEC cannot give. Quote ν and h with caution.

Controller tracking (PI)



Top: setpoint, TEC thermistor and T1 over the settled block. Bottom: PWM commanded.

P (s)	B/Tb	Lag (s)	PWM
480	0.99	12.0	−47 to 83
240	0.96	11.4	−94 to 101
100	0.86	9.9	−225 to 152

- B/Tb: TEC amplitude as a fraction of the 16 °C setpoint amplitude.
- The TEC thermistor lags the setpoint by about 10 to 12 s at every period.
- Hold step: TEC reached 39.5 °C after 45 s with +0.03 °C overshoot.
- No sample at ±255 PWM (0).

The controller follows the sine closely at 480 and 240 s; at 100 s the TEC amplitude falls to 86 % of the setpoint's, so the rod is driven a little less than the 16 °C asked for.

Conclusions, limits and next steps

What the run shows

- $\kappa = 9.12 \pm 0.09 \times 10^{-5}$ m²/s from three periods; they are consistent with one value ($\chi^2 = 3.6$, 2 dof).
- Within 0.5 to 1.7 σ of Kapil's PI fits at each period: same rod, same method.
- 32 % above the 6061-T6 handbook value (k ≈ 220 against 167 W/(m K) with the handbook ρ and cp). Kapil's data show it too, so it is not an accident of this run.
- Only 240 s has five settled cycles; 480 s and 100 s are supplementary.

Limits and next steps

- ν and h: ν is a small difference of two close numbers, and the mean-profile slope disagrees (3.7 to 4.2 against 4.7 to 5.0 1/m). Quote them with caution.
- T5 needed masking in the 480 s step; the settling tolerances are this analysis's own.
- Not covered by this file: open-loop and P-only comparisons, the raw sinusoidal PWM trials (S18.3, S18.4), gain tuning at 200 s and 500 s, the P-versus-PI simulation.
- Next: repeat the 480 s and 100 s steps (or run them longer) to reach five settled cycles, and test the offset from the handbook, for example with a second rod of known alloy.

The method and the controller worked; the open question is the 32 % offset from the handbook diffusivity, which this run reproduces rather than explains.

Files, reproduction and AI assistance

Reproduce

```
python3 python/sammy_module8_angstrom_analysis.py \
    data/module_08_sammy/sammy_angstrom_live_20261110_164057.csv

python3 python/build_sammy_module8_slides.py \
    data/module_08_sammy/sammy_angstrom_live_20261110_164057.csv
```

Where things are

- data/module_08_sammy/: the run CSV, its events, meta and notes, the mask file, analysis_20261110_164057/ (tables, JSON, report).
- docs/figures/module_08_sammy/ and docs/images/: figures and photos.
- docs/sammy_module_08_guide/: the written summary and this deck.
- python/: sammy_module8_angstrom_gui.py (acquisition), sammy_module8_angstrom_analysis.py, build_sammy_module8_slides.py.

AI assistance

- The acquisition-GUI changes, the analysis script, the figures, these slides and the written summary were produced with GitHub Copilot assistance.
- They follow Kapil Naidu's angstrom_fit.py method and the Module 8 study guide, and were checked against his fitted values.
- Kapil's acquisition code, analysis scripts and data are unchanged; everything here is an addition.

Every number on these slides is read from the analysis tables; rerunning the two commands rebuilds the tables, figures, summary and deck from the run CSV.